

MIT 18.06 Linear Algebra Cheat Sheet: Chapters 1–3

Based on Gilbert Strang, 5th Edition

Chapter 1: Introduction to Vectors

1.1 Vector Operations

- **Linear Combination:** $cv + dw$ (adding cv to dw).
- **Geometry:** Combinations of u, v, w typically fill a line, a plane, or the whole space $\mathbb{R}^3$.
- **Notation:** $v = (v_1, v_2)$ is a column vector written in row form to save space. v^T is the actual row vector.

1.2 Dot Products and Length

- **Dot Product:** $v \cdot w = v_1w_1 + v_2w_2 + \cdots + v_nw_n$.
- **Length (Norm):** $\|v\| = \sqrt{v \cdot v} = \sqrt{v_1^2 + \cdots + v_n^2}$.
- **Unit Vector:** $u = v/\|v\|$ has length 1. Standard unit vectors: $i = (1, 0), j = (0, 1)$.
- **Angle Formula:** $\cos \theta = \frac{v \cdot w}{\|v\|\|w\|}$.
- **Orthogonality:** v and w are perpendicular if $v \cdot w = 0$.
- **Schwarz Inequality:** $|v \cdot w| \leq \|v\|\|w\|$.
- **Triangle Inequality:** $\|v + w\| \leq \|v\| + \|w\|$.

1.3 Matrices

- **Matrix-Vector Product:** Ax is a linear combination of the columns of A .
- **Independence:** Columns are independent if $Ax = 0$ has only the solution $x = 0$.

Chapter 2: Solving Linear Equations

2.1 Row and Column Pictures

- **Row Picture:** Each row of $Ax = b$ is an equation for a line ($n = 2$), plane ($n = 3$), or hyperplane ($n > 3$).
- **Column Picture:** Finds weights x_i to combine column vectors to produce b : $x_1a_1 + \cdots + x_na_n = b$.

2.2-2.3 Elimination

- **Mechanism:** Uses multipliers $\ell_{ij} = (\text{entry to eliminate})/(\text{pivot})$ to produce zeros below pivots.
- **Pivots:** The first non-zero entry in each row during elimination. Matrix is singular if a pivot is 0 and cannot be exchanged.
- **Elimination Matrices:** E_{21} subtracts a multiple of row 1 from row 2.

2.4-2.5 Matrix Algebra and Inverses

- **Matrix Product AB :** Entry $C_{ij} = (\text{row } i \text{ of } A) \cdot (\text{column } j \text{ of } B)$.
- **Rules:** $A(BC) = (AB)C$ (Associative); $A(B + C) = AB + AC$ (Distributive); $AB \neq BA$ usually.
- **Inverse A^{-1} :** Exists if elimination produces n pivots. $AA^{-1} = I = A^{-1}A$.
- **Properties:** $(AB)^{-1} = B^{-1}A^{-1}$.
- **Gauss-Jordan:** Solves $[A \mid I] \rightarrow [I \mid A^{-1}]$ by row operations.

2.6 Factorization $A = LU$

- L is lower triangular (multipliers below the diagonal); U is upper triangular (pivots on the diagonal).
- $A = LDU$ separates pivots into a diagonal matrix D .

2.7 Transposes and Permutations

- **Transpose:** $A_{ij}^T = A_{ji}$. Rules: $(AB)^T = B^T A^T$.
- **Symmetric Matrix:** $S^T = S$. $(S^{-1})^T = S^{-1}$.
- **Permutations:** P exchanges rows. $P^T = P^{-1}$.

Chapter 3: Vector Spaces and Subspaces

3.1-3.2 Spaces and Nullspaces

- **Vector Space:** Set of vectors closed under addition and scalar multiplication.
- **Subspaces:** Must contain the zero vector 0. Example: Plane through origin.
- **Column Space $C(A)$:** All combinations Ax . Solvable if $b \in C(A)$.
- **Nullspace $N(A)$:** All solutions to $Ax = 0$.

3.3 The Complete Solution to $Ax = b$

- **Rank r :** Number of pivots. Dimension of column space.
- **Reduced Row Echelon Form (R):** r pivots are 1s, zeros above and below pivots.

- **Variables:** r pivot variables and $n - r$ free variables.
- **Complete Solution:** $x = x_p + x_n$. One particular solution (x_p) plus any nullspace solution (x_n).

3.4 Independence, Basis, and Dimension

- **Independent:** Only 0 combination gives 0. Columns are independent if $N(A) = \{0\}$.
- **Span:** Vectors span a space if their combinations fill it.
- **Basis:** Linearly independent vectors that span the space.
- **Dimension:** Number of vectors in any basis.

3.5 The Four Fundamental Subspaces

- **Column Space $C(A)$:** $\dim r$, subspace of $\mathbb{R}^m$.
- **Row Space $C(A^T)$:** $\dim r$, subspace of $\mathbb{R}^n$.
- **Nullspace $N(A)$:** $\dim n - r$, subspace of $\mathbb{R}^n$.
- **Left Nullspace $N(A^T)$:** $\dim m - r$, subspace of $\mathbb{R}^m$.
- **Fundamental Theorem I:** Row and column spaces both have dimension r .